Activity

Group members:

Inference with Poisson regression

We are interested in analyzing the number of articles published by biochemistry PhD students. The data contains the following variables:

- `art`: articles published in last three years of Ph.D.
- `fem`: gender (recorded as men or women)
- `mar`: marital status (recorded as married or single)
- `kid5`: number of children under age six
- `phd`: prestige of Ph.D. program
- `ment`: articles published by their mentor in last three years

Here is the model output for a Poisson regression model:

```
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.304617   0.102981   2.958   0.0031 **
femWomen    -0.224594   0.054613  -4.112 3.92e-05 ***
marMarried   0.155243   0.061374   2.529   0.0114 *
kid5        -0.184883   0.040127  -4.607 4.08e-06 ***
phd          0.012823   0.026397   0.486   0.6271
ment         0.025543   0.002006  12.733  < 2e-16 ***

    Null deviance: 1817.4  on 914  degrees of freedom
Residual deviance: 1634.4  on 909  degrees of freedom
```

Questions

1. Using the R output, perform a goodness of fit test to assess whether this Poisson regression model is a good fit to the data.